Nyra MES Core — data structure reference

2026-10-05 · database postgres · nyra-mes-core · structure and meaning only. No live rows. Use this as a printed map of where data lives.

1. How the database is organised

One PostgreSQL database holds every industry. Each plant is a **tenant**. GR Carbons is **GR_CARBONS**, workspace **GR_CARBONS_ACTIVATED_CARBON**. Shop-floor identity is the inward order **ORD-AC-***. The system still creates a hidden PO and lot for the ontology.

Schema	What it holds
core	Tenancy, identity, and cross-tenant platform metadata
template	Industry templates (AUTOMOBILE_OEM, FOUNDRY, TEXTILE, …)
mst	Master data — tenant-scoped
trn	Transactional manufacturing documents — tenant-scoped
exec	Shop-floor execution events — tenant-scoped
qms	Quality management — tenant-scoped
kg	Knowledge graph staging (triples export/import)
ont	Future: ont.class, ont.relationship, ont.entity_mapping — no tables required at bootstrap

2. Table catalogue

Table	What it stores
core.app_user	Login identity (email, Cognito sub). Passwords are not stored here.
core.capability	Platform capability catalog (industry-agnostic). Referenced by template.industry_capability.
core.project_workspace	Tenant-scoped project/plant workspaces. Future industry data may scope to workspace_id.
core.tenant	One row per plant / customer. GR Carbons is tenant_code GR_CARBONS.
core.user_project_access	Maps users to project workspaces with MES role codes.
core.user_tenant_access	Which MES role a user has on a tenant (RMS, Activation, Admin, …).
template.industry_capability	Maps industry templates to enabled platform capabilities.
template.industry_template	Catalog of industry templates. New industries = new rows (no schema migration).
template.raw_material_parameter_profile	Activated Carbon raw-material parameter bands — shared across AC_STANDARD_PROCESS (no per-material routes).
mst.customer	End customers. Also the buyer on inward (supplier receipt) POs.
mst.equipment	Physical equipment linked to a resource.
mst.modbus_register_map	Modbus register map for live kiln telemetry.
mst.operation_resource_requirement	Which resource an operation needs.
mst.parameter_business_mapping	Maps a sensor parameter to a business reading name.
mst.product	Finished / intermediate grades (FG-AC-*).
mst.resource	Kilns, rotaries, machines, labor, material resources.
mst.routing	Process route for a product (AC standard process).
mst.routing_operation	Ordered steps on a route (kiln feed, destoning, screening, …).
mst.routing_operation_parameter	Workspace-scoped operation parameters (limits, UOM, capture frequency).
mst.sensor_device	Kiln / plant sensor device register.
mst.sensor_parameter	Named parameters on a sensor device (temperature, etc.).
mst.supplier	Raw material vendors — separate from end customers (mst.customer)
trn.customer_order	Inward receipts (ORD-AC-*) and customer sales POs (CUST_PO_*). Shop-floor RM detail is in metadata.
trn.customer_order_line	Product + quantity lines on an inward or sales order.
trn.dispatch	Shipment dispatch records (Activated Carbon and future industries). Distinct from legacy oemauto dispatch.
trn.feasibility_check	Feasibility header. The UI gate is skipped; rows may still exist for ontology.

Table	What it stores
trn.feasibility_check_detail	One line per feasibility check item.
trn.production_lot	Lot created for the ontology (LOT-AC-*). Kiln / destoning / post-activation detail is in metadata.
trn.production_order	Internal PO created for the ontology when inward is received. Not shown on the shop floor.
trn.sensor_reading	One telemetry snapshot from a sensor device.
trn.sensor_reading_value	Individual values inside a telemetry snapshot.
exec.operation_execution	One shop-floor run of a routing step on a lot.
exec.parameter_reading	Numeric / text reading captured on an execution.
qms.quality_check	QC header for a lot (lab, destoning, kiln output, dispatch).
qms.quality_result	Measured characteristics for a QC header.
kg.triple_store_staging	Ontology triples: subject / predicate / object for inward and lot URIs.

3. The metadata JSON column — not a file

There is no separate JSON file. Several tables have a column named **metadata**. That column is a notebook attached to the same row. Fixed columns hold identity (order code, lot code, status, product, quantity). The notebook holds the process the plant typed on the screens.

trn.customer_order — inward ORD-AC-*

Key	What it stores
inward_receipt	DC number, supplier KG, our KG from gate receipt.
initial_qc / inbound_execution.IN-LAB	Lab moisture, dust, AD, approve or deny.
inward_bags	RMS inward bags (RI_BAG_*), weight, fed-to-crusher status.
bag_reconciliation	Bag total vs our inward weight.
inbound_execution.IN-RMS-BAG	RMS inward step status and bag totals.
inbound_execution.IN-CRUSHER / crusher_output	Rotary, mesh %, moisture, dust.
grade_separation / IN-RM-OUT	Outward bags (RO_BAG_*), rotary, grade, KG.
released_lots	Lots and KG released to production.
flow	RAW_MATERIAL_INWARD for receipts.

trn.customer_order — sales CUST_PO_*

Key	What it stores
flow	CUSTOMER_SALES for a customer PO.
ordered_kg / dispatched_kg	Ordered and already shipped weight.
product_id	Grade the customer ordered.

trn.production_lot — LOT-AC-*

Key	What it stores
source_inward_order_id	Which inward the lot came from.
kiln_resource_code / kiln_loaded_kg	Which kiln and how much was charged.
kiln_feed_events	Each load: bag code, KG, time.
crusher_outward_bags	RO bags attached when the lot was released.
destoning_output_bags	Destoning bags, grade, destination, In Stock QC.
post-activation bag lists	Screening / crushing / demagnetisation / blending bags.

kg.triple_store_staging

Key	What it stores
subject_uri	urn:nyra:ac:inward:{ORD-*} or urn:nyra:ac:lot:{LOT-*}. Do not replace.
predicate_uri	The fact (received, lab approved, bag created, charged, dispatched).
object_uri / object_literal	The related thing or the value.

4. Columns by table

Print this section as the desk reference. PK = primary key. FK = link to another table.

core.app_user

Login identity (email, Cognito sub). Passwords are not stored here.

Primary key: user_id.

Column	Type	Key	What it stores
user_id	UUID	PK, required	Link to core.app_user.
cognito_sub	VARCHAR(128)	—	Cognito login identifier.
email	VARCHAR(320)	required	User login email.
display_name	VARCHAR(256)	—	Name shown in the UI.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

core.capability

Platform capability catalog (industry-agnostic). Referenced by template.industry_capability.

Primary key: capability_id.

Column	Type	Key	What it stores
capability_id	UUID	PK, required	Field stored on this table.
capability_code	VARCHAR(128)	required	Field stored on this table.
capability_name	VARCHAR(256)	required	Field stored on this table.
description	TEXT	—	Field stored on this table.
category	VARCHAR(64)	—	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

core.project_workspace

Tenant-scoped project/plant workspaces. Future industry data may scope to workspace_id.

Primary key: workspace_id.

Links: tenant_id → core.tenant.

Column	Type	Key	What it stores
workspace_id	UUID	PK, required	Which workspace / plant area this row belongs to.
tenant_id	UUID	FK, required	Which plant this row belongs to.
workspace_code	VARCHAR(64)	required	Readable workspace code (GR_CARBONS_ACTIVATED_CARBON).
workspace_name	VARCHAR(256)	required	Field stored on this table.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

core.tenant

One row per plant / customer. GR Carbons is tenant_code GR_CARBONS.

Primary key: tenant_id.

Links: industry_template_id → template.industry_template.

Column	Type	Key	What it stores
tenant_id	UUID	PK, required	Which plant this row belongs to.
tenant_code	VARCHAR(64)	required	Stable plant code (GR_CARBONS).
tenant_name	VARCHAR(256)	required	Display name of the plant.
industry_template_id	UUID	FK, required	Link to the industry pack.
status	VARCHAR(32)	required	Business status of this row.
timezone	VARCHAR(64)	required	Plant timezone.
metadata	JSONB	required	Tenant extras. Not shop-floor capture.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

core.user_project_access

Maps users to project workspaces with MES role codes.

Primary key: access_id.

Links: user_id → core.app_user; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
access_id	UUID	PK, required	Field stored on this table.
user_id	UUID	FK, required	Link to core.app_user.
workspace_id	UUID	FK, required	Which workspace / plant area this row belongs to.
role_code	VARCHAR(64)	required	MES role (MES_RMS, MES_Activation, MES_Admin, …).
is_default	BOOLEAN	required	Default tenant or workspace for this user.
granted_at	TIMESTAMP	required	Field stored on this table.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

core.user_tenant_access

Which MES role a user has on a tenant (RMS, Activation, Admin, …).

Primary key: access_id.

Links: tenant_id → core.tenant; user_id → core.app_user.

Column	Type	Key	What it stores
access_id	UUID	PK, required	Field stored on this table.
user_id	UUID	FK, required	Link to core.app_user.
tenant_id	UUID	FK, required	Which plant this row belongs to.
role_code	VARCHAR(64)	required	MES role (MES_RMS, MES_Activation, MES_Admin, …).
is_default	BOOLEAN	required	Default tenant or workspace for this user.
granted_at	TIMESTAMP	required	Field stored on this table.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

template.industry_capability

Maps industry templates to enabled platform capabilities.

Primary key: industry_capability_id.

Links: capability_id → core.capability; industry_template_id → template.industry_template.

Column	Type	Key	What it stores
industry_capability_id	UUID	PK, required	Field stored on this table.
industry_template_id	UUID	FK, required	Link to the industry pack.
capability_id	UUID	FK, required	Field stored on this table.
display_order	INTEGER	required	Field stored on this table.

Column	Type	Key	What it stores
is_required	BOOLEAN	required	Field stored on this table.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

template.industry_template

Catalog of industry templates. New industries = new rows (no schema migration).

Primary key: industry_template_id.

Column	Type	Key	What it stores
industry_template_id	UUID	PK, required	Link to the industry pack.
template_code	VARCHAR(64)	required	Field stored on this table.
template_name	VARCHAR(256)	required	Field stored on this table.
description	TEXT	—	Field stored on this table.
template_version	VARCHAR(32)	required	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
module_code	VARCHAR(128)	—	Field stored on this table.
landing_page	VARCHAR(512)	—	Field stored on this table.
ui_module_name	VARCHAR(128)	—	Field stored on this table.
api_module_name	VARCHAR(128)	—	Field stored on this table.
icon_name	VARCHAR(64)	—	Field stored on this table.
dashboard_component	VARCHAR(128)	—	Field stored on this table.
display_order	INTEGER	required	Field stored on this table.

template.raw_material_parameter_profile

Activated Carbon raw-material parameter bands — shared across AC_STANDARD_PROCESS (no per-material routes).

Primary key: id.

Column	Type	Key	What it stores
id	UUID	PK, required	Field stored on this table.
industry_template_code	VARCHAR(64)	required	Industry pack code (ACTIVATED_CARBON).
raw_material_type	VARCHAR(128)	required	Coconut shell, wood, etc.
profile_code	VARCHAR(128)	required	Parameter-profile code.
profile_name	VARCHAR(256)	—	Parameter-profile name.
parameter_code	VARCHAR(64)	required	Short code of a captured parameter.
parameter_name	VARCHAR(256)	—	Display name of a captured parameter.
min_value	NUMERIC(18, 6)	—	Lower limit.
max_value	NUMERIC(18, 6)	—	Upper limit.
unit	VARCHAR(32)	—	Unit for the parameter.
expected_yield_percent	NUMERIC(8, 4)	—	Expected yield for this material.
expected_ctc_min	NUMERIC(18, 6)	—	Field stored on this table.
expected_ctc_max	NUMERIC(18, 6)	—	Field stored on this table.
expected_density_min	NUMERIC(18, 6)	—	Field stored on this table.
expected_density_max	NUMERIC(18, 6)	—	Field stored on this table.

Column	Type	Key	What it stores
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

mst.customer

End customers. Also the buyer on inward (supplier receipt) POs.

Primary key: customer_id.

Links: tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
customer_id	UUID	PK, required	Link to mst.customer.
tenant_id	UUID	FK, required	Which plant this row belongs to.
customer_code	VARCHAR(64)	required	Customer code.
customer_name	VARCHAR(256)	required	Customer display name.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

mst.equipment

Physical equipment linked to a resource.

Primary key: equipment_id.

Links: resource_id → mst.resource; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
equipment_id	UUID	PK, required	Link to mst.equipment.
tenant_id	UUID	FK, required	Which plant this row belongs to.
equipment_code	VARCHAR(64)	required	Equipment code.
equipment_name	VARCHAR(256)	required	Equipment display name.
resource_id	UUID	FK	Link to mst.resource.
equipment_type	VARCHAR(64)	—	Equipment class.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

mst.modbus_register_map

Modbus register map for live kiln telemetry.

Primary key: id.

Links: device_id → mst.sensor_device; parameter_id → mst.sensor_parameter.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
device_id	BIGINT	FK, required	Sensor device id.
parameter_id	BIGINT	FK, required	Field stored on this table.
slave_id	INTEGER	required	Field stored on this table.
function_code	INTEGER	required	Field stored on this table.

Column	Type	Key	What it stores
register_address	INTEGER	required	Modbus register address.
register_count	INTEGER	required	Field stored on this table.
data_type	VARCHAR(50)	required	Expected value type (number, text, …).
byte_order	VARCHAR(50)	required	Field stored on this table.
scale_factor	NUMERIC	required	Field stored on this table.
offset_value	NUMERIC	required	Field stored on this table.
simulation_min	NUMERIC	—	Field stored on this table.
simulation_max	NUMERIC	—	Field stored on this table.
polling_interval_sec	INTEGER	required	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

mst.operation_resource_requirement

Which resource an operation needs.

Primary key: requirement_id.

Links: resource_id → mst.resource; routing_operation_id → mst.routing_operation; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
requirement_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
routing_operation_id	UUID	FK, required	Link to one step on a route.
resource_id	UUID	FK, required	Link to mst.resource.
requirement_type	VARCHAR(32)	required	Field stored on this table.
quantity	NUMERIC(18, 4)	required	Quantity on this line.
uom	VARCHAR(16)	required	Unit of measure (KG, EA).
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

mst.parameter_business_mapping

Maps a sensor parameter to a business reading name.

Primary key: id.

Links: device_id → mst.sensor_device; parameter_id → mst.sensor_parameter.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
industry_code	VARCHAR(100)	required	Field stored on this table.
device_id	BIGINT	FK, required	Sensor device id.
parameter_id	BIGINT	FK, required	Field stored on this table.
target_entity	VARCHAR(100)	required	Field stored on this table.
target_field	VARCHAR(100)	required	Field stored on this table.
process_step_code	VARCHAR(100)	—	Field stored on this table.
work_center_code	VARCHAR(100)	—	Field stored on this table.
equipment_code	VARCHAR(100)	—	Equipment code.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.

Column	Type	Key	What it stores
updated_at	TIMESTAMP	required	When the row was last changed.

mst.product

Finished / intermediate grades (FG-AC-*).

Primary key: product_id.

Links: tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
product_id	UUID	PK, required	Link to mst.product.
tenant_id	UUID	FK, required	Which plant this row belongs to.
product_code	VARCHAR(64)	required	Product / grade code (FG-AC-*).
product_name	VARCHAR(256)	required	Product / grade name.
product_type	VARCHAR(64)	—	Product class (finished, raw, intermediate).
uom	VARCHAR(16)	required	Unit of measure (KG, EA).
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Whether this product is a raw-material type, inbound operations, parameters.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

mst.resource

Kilns, rotaries, machines, labor, material resources.

Primary key: resource_id.

Links: tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
resource_id	UUID	PK, required	Link to mst.resource.
tenant_id	UUID	FK, required	Which plant this row belongs to.
resource_code	VARCHAR(64)	required	Resource code (kiln, rotary, …).
resource_name	VARCHAR(256)	required	Resource display name.
resource_type	VARCHAR(64)	required	MACHINE, LABOR, TOOL, MATERIAL, WORKCELL, OTHER.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	In-use count and kiln / rotary extras.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

mst.routing

Process route for a product (AC standard process).

Primary key: routing_id.

Links: product_id → mst.product; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
routing_id	UUID	PK, required	Link to mst.routing.
tenant_id	UUID	FK, required	Which plant this row belongs to.
routing_code	VARCHAR(64)	required	Route code.
product_id	UUID	FK	Link to mst.product.
routing_version	VARCHAR(32)	required	Route version.
process_family	VARCHAR(64)	—	Family of the route (activation, inbound, …).

Column	Type	Key	What it stores
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
metadata	JSONB	required	Route extras and industry flags.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
industry_template_code	VARCHAR(64)	—	Industry pack code (ACTIVATED_CARBON).
source_template_code	VARCHAR(64)	—	Field stored on this table.
is_template_route	BOOLEAN	required	Field stored on this table.
is_template_copy	BOOLEAN	required	Field stored on this table.
is_customized	BOOLEAN	required	Field stored on this table.

mst.routing_operation

Ordered steps on a route (kiln feed, destoning, screening, …).

Primary key: routing_operation_id.

Links: routing_id → mst.routing; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
routing_operation_id	UUID	PK, required	Link to one step on a route.
tenant_id	UUID	FK, required	Which plant this row belongs to.
routing_id	UUID	FK, required	Link to mst.routing.
operation_code	VARCHAR(64)	required	Step code (IN-LAB, KILN-FEED, DESTONING, …).
operation_sequence	INTEGER	required	Order of the step on the route.
operation_name	VARCHAR(256)	required	Step display name.
standard_cycle_min	NUMERIC(12, 4)	—	Standard cycle time in minutes.
metadata	JSONB	required	process_step and whether the step is mandatory.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
industry_template_code	VARCHAR(64)	—	Industry pack code (ACTIVATED_CARBON).
source_template_operation_code	VARCHAR(64)	—	Field stored on this table.
operation_seq	INTEGER	—	Field stored on this table.
operation_type	VARCHAR(64)	—	Field stored on this table.
is_quality_gate	BOOLEAN	required	Field stored on this table.
is_feasibility_gate	BOOLEAN	required	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
is_customized	BOOLEAN	required	Field stored on this table.

mst.routing_operation_parameter

Workspace-scoped operation parameters (limits, UOM, capture frequency).

Primary key: id.

Links: routing_operation_id → mst.routing_operation; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
routing_operation_id	UUID	FK, required	Link to one step on a route.
parameter_code	VARCHAR(64)	—	Short code of a captured parameter.

Column	Type	Key	What it stores
parameter_name	VARCHAR(256)	—	Display name of a captured parameter.
data_type	VARCHAR(32)	—	Expected value type (number, text, …).
unit	VARCHAR(32)	—	Unit for the parameter.
min_value	NUMERIC(18, 6)	—	Lower limit.
max_value	NUMERIC(18, 6)	—	Upper limit.
is_mandatory	BOOLEAN	required	Whether the plant must fill this.
reading_frequency	VARCHAR(64)	—	Field stored on this table.
display_order	INTEGER	—	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

mst.sensor_device

Kiln / plant sensor device register.

Primary key: id.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
device_code	VARCHAR(100)	required	Sensor device code.
device_name	VARCHAR(200)	required	Field stored on this table.
industry_code	VARCHAR(100)	required	Field stored on this table.
site_code	VARCHAR(100)	—	Field stored on this table.
area_code	VARCHAR(100)	—	Field stored on this table.
work_center_code	VARCHAR(100)	—	Field stored on this table.
equipment_code	VARCHAR(100)	—	Equipment code.
protocol	VARCHAR(50)	required	Field stored on this table.
connection_type	VARCHAR(50)	required	Field stored on this table.
ip_address	VARCHAR(100)	—	Field stored on this table.
port	INTEGER	required	Field stored on this table.
serial_port	VARCHAR(100)	—	Field stored on this table.
baud_rate	INTEGER	—	Field stored on this table.
parity	VARCHAR(20)	—	Field stored on this table.
stop_bits	INTEGER	—	Field stored on this table.
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

mst.sensor_parameter

Named parameters on a sensor device (temperature, etc.).

Primary key: id.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
parameter_code	VARCHAR(100)	required	Short code of a captured parameter.
parameter_name	VARCHAR(200)	required	Display name of a captured parameter.
data_type	VARCHAR(50)	required	Expected value type (number, text, …).
uom	VARCHAR(50)	—	Unit of measure (KG, EA).
normal_min	NUMERIC	—	Field stored on this table.
normal_max	NUMERIC	—	Field stored on this table.

Column	Type	Key	What it stores
is_active	BOOLEAN	required	Whether this master / catalog row is still in use.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

mst.supplier

Raw material vendors — separate from end customers (mst.customer)

Primary key: supplier_id.

Links: tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
supplier_id	UUID	PK, required	Link to mst.supplier.
tenant_id	UUID	FK, required	Which plant this row belongs to.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
supplier_code	VARCHAR(64)	required	Supplier code.
supplier_name	VARCHAR(256)	required	Supplier display name.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

trn.customer_order

Inward receipts (ORD-AC-*) and customer sales POs (CUST_PO_*). Shop-floor RM detail is in metadata.

Primary key: customer_order_id.

Links: customer_id → mst.customer; supplier_id → mst.supplier; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
customer_order_id	UUID	PK, required	Link to trn.customer_order.
tenant_id	UUID	FK, required	Which plant this row belongs to.
order_code	VARCHAR(64)	required	Inward ORD-AC-* or sales CUST_PO_*.
customer_id	UUID	FK	Link to mst.customer.
order_date	DATE	required	Document date.
status	VARCHAR(32)	required	OPEN, QC_PASSED, RELEASED, CLOSED, etc. Inward vs sales is also in metadata.flow.
metadata	JSONB	required	Gate receipt, lab, RMS bags, crusher, outward, release. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
supplier_id	UUID	FK	Link to mst.supplier.

trn.customer_order_line

Product + quantity lines on an inward or sales order.

Primary key: order_line_id.

Links: order_id → trn.customer_order; product_id → mst.product; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
order_line_id	UUID	PK, required	Link to trn.customer_order_line.
tenant_id	UUID	FK, required	Which plant this row belongs to.
order_id	UUID	FK, required	Link to trn.customer_order.
line_code	VARCHAR(64)	required	Order line code.
product_id	UUID	FK, required	Link to mst.product.

Column	Type	Key	What it stores
quantity	NUMERIC(18, 4)	required	Quantity on this line.
uom	VARCHAR(16)	required	Unit of measure (KG, EA).
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Line extras (grade, remaining qty) when used.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

trn.dispatch

Shipment dispatch records (Activated Carbon and future industries). Distinct from legacy oemauto dispatch.

Primary key: id.

Links: customer_order_id → trn.customer_order; production_lot_id → trn.production_lot; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.
customer_order_id	UUID	FK	Link to trn.customer_order.
production_lot_id	UUID	FK	Link to the lot.
finished_goods_lot_no	VARCHAR(128)	—	Finished-goods lot shown on dispatch.
grade	VARCHAR(64)	—	Dispatched grade.
bag_count	NUMERIC(18, 4)	—	Number of bags dispatched.
total_weight	NUMERIC(18, 4)	—	Dispatched weight.
vehicle_no	VARCHAR(64)	—	Vehicle number.
dispatch_date	DATE	—	Dispatch date.
invoice_ref	VARCHAR(128)	—	Invoice reference.
destination	TEXT	—	Ship-to destination.
status	VARCHAR(32)	required	Business status of this row.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.

trn.feasibility_check

Feasibility header. The UI gate is skipped; rows may still exist for ontology.

Primary key: feasibility_check_id.

Links: production_lot_id → trn.production_lot; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
feasibility_check_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
check_code	VARCHAR(64)	required	Feasibility or QC document code.
production_lot_id	UUID	FK, required	Link to the lot.
overall_status	VARCHAR(32)	required	Feasibility result (PENDING, OK, BLOCKED, WAIVED).
checked_at	TIMESTAMP	—	When feasibility was recorded.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

trn.feasibility_check_detail

One line per feasibility check item.

Primary key: feasibility_detail_id.

Links: feasibility_check_id → trn.feasibility_check; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
feasibility_detail_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
feasibility_check_id	UUID	FK, required	Field stored on this table.
check_key	VARCHAR(128)	required	Feasibility item key.
check_label	VARCHAR(256)	—	Feasibility item label.
status	VARCHAR(32)	required	Business status of this row.
message	TEXT	—	Feasibility message.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

trn.production_lot

Lot created for the ontology (LOT-AC-*). Kiln / destoning / post-activation detail is in metadata.

Primary key: production_lot_id.

Links: production_order_id → trn.production_order; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
production_lot_id	UUID	PK, required	Link to the lot.
tenant_id	UUID	FK, required	Which plant this row belongs to.
lot_code	VARCHAR(64)	required	Lot code (LOT-AC-*). Ontology identity; shop floor uses the inward order.
production_order_id	UUID	FK, required	Link to the internal production order.
lot_quantity	NUMERIC(18, 4)	required	Quantity on the lot.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Kiln feed, destoning bags, post-activation bags, In Stock. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

trn.production_order

Internal PO created for the ontology when inward is received. Not shown on the shop floor.

Primary key: production_order_id.

Links: order_line_id → trn.customer_order_line; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
production_order_id	UUID	PK, required	Link to the internal production order.
tenant_id	UUID	FK, required	Which plant this row belongs to.
production_order_code	VARCHAR(64)	required	Internal PO code (PO-AC-*). Hidden on the shop floor.
order_line_id	UUID	FK, required	Link to trn.customer_order_line.
planned_quantity	NUMERIC(18, 4)	required	Planned quantity on the internal PO.
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.

Column	Type	Key	What it stores
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

trn.sensor_reading

One telemetry snapshot from a sensor device.

Primary key: id.

Links: device_id → mst.sensor_device.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
device_id	BIGINT	FK, required	Sensor device id.
reading_timestamp	TIMESTAMP	required	Field stored on this table.
source	VARCHAR(50)	required	Field stored on this table.
payload_format	VARCHAR(50)	required	Field stored on this table.
protocol	VARCHAR(50)	required	Field stored on this table.
quality_status	VARCHAR(50)	required	Field stored on this table.
raw_payload	JSONB	—	Field stored on this table.
created_at	TIMESTAMP	required	When the row was created.

trn.sensor_reading_value

Individual values inside a telemetry snapshot.

Primary key: id.

Links: parameter_id → mst.sensor_parameter; sensor_reading_id → trn.sensor_reading.

Column	Type	Key	What it stores
id	BIGINT	PK, required	Field stored on this table.
sensor_reading_id	BIGINT	FK, required	Field stored on this table.
parameter_id	BIGINT	FK, required	Field stored on this table.
parameter_code	VARCHAR(100)	required	Short code of a captured parameter.
value_numeric	NUMERIC	—	Field stored on this table.
value_text	VARCHAR(200)	—	Field stored on this table.
uom	VARCHAR(50)	—	Unit of measure (KG, EA).
quality	VARCHAR(50)	required	Field stored on this table.
slave_id	INTEGER	—	Field stored on this table.
function_code	INTEGER	—	Field stored on this table.
register_address	INTEGER	—	Modbus register address.
raw_value	VARCHAR(200)	—	Field stored on this table.
created_at	TIMESTAMP	required	When the row was created.

exec.operation_execution

One shop-floor run of a routing step on a lot.

Primary key: execution_id.

Links: equipment_id → mst.equipment; production_lot_id → trn.production_lot; routing_operation_id → mst.routing_operation; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
execution_id	UUID	PK, required	Link to exec.operation_execution.
tenant_id	UUID	FK, required	Which plant this row belongs to.
execution_code	VARCHAR(64)	required	Shop-floor run code.
production_lot_id	UUID	FK, required	Link to the lot.
routing_operation_id	UUID	FK, required	Link to one step on a route.

Column	Type	Key	What it stores
equipment_id	UUID	FK	Link to mst.equipment.
status	VARCHAR(32)	required	Business status of this row.
started_at	TIMESTAMP	—	When the run started.
completed_at	TIMESTAMP	—	When the run finished.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

exec.parameter_reading

Numeric / text reading captured on an execution.

Primary key: parameter_reading_id.

Links: execution_id → exec.operation_execution; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
parameter_reading_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
execution_id	UUID	FK, required	Link to exec.operation_execution.
parameter_code	VARCHAR(128)	required	Short code of a captured parameter.
parameter_name	VARCHAR(256)	—	Display name of a captured parameter.
reading_value	NUMERIC(18, 6)	—	Numeric reading.
reading_text	TEXT	—	Text reading.
uom	VARCHAR(16)	—	Unit of measure (KG, EA).
recorded_at	TIMESTAMP	required	When the reading was taken.
metadata	JSONB	required	Flexible shop-floor notebook for this row. Not a separate file. See section 3.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

qms.quality_check

QC header for a lot (lab, destoning, kiln output, dispatch).

Primary key: quality_check_id.

Links: production_lot_id → trn.production_lot; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
quality_check_id	UUID	PK, required	Link to qms.quality_check.
tenant_id	UUID	FK, required	Which plant this row belongs to.
check_code	VARCHAR(64)	required	Feasibility or QC document code.
production_lot_id	UUID	FK, required	Link to the lot.
check_type	VARCHAR(64)	required	QC type (IN_PROCESS, DISPATCH, …).
status	VARCHAR(32)	required	Business status of this row.
metadata	JSONB	required	Dispatch / bag QC header extras (vehicle, grade, lines).
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

qms.quality_result

Measured characteristics for a QC header.

Primary key: quality_result_id.

Links: quality_check_id → qms.quality_check; tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
quality_result_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
quality_check_id	UUID	FK, required	Link to qms.quality_check.
characteristic_code	VARCHAR(128)	required	QC characteristic code.
characteristic_name	VARCHAR(256)	—	QC characteristic name.
measured_value	NUMERIC(18, 6)	—	Numeric QC result.
measured_text	TEXT	—	Text QC result.
uom	VARCHAR(16)	—	Unit of measure (KG, EA).
result_status	VARCHAR(32)	required	PASS / FAIL / etc.
recorded_at	TIMESTAMP	required	When the reading was taken.
metadata	JSONB	required	Per-characteristic extras or bag destinations.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.

kg.triple_store_staging

Ontology triples: subject / predicate / object for inward and lot URIs.

Primary key: triple_id.

Links: tenant_id → core.tenant; workspace_id → core.project_workspace.

Column	Type	Key	What it stores
triple_id	UUID	PK, required	Field stored on this table.
tenant_id	UUID	FK, required	Which plant this row belongs to.
subject_uri	TEXT	required	Ontology subject, e.g. urn:nyra:ac:inward:ORD-AC-20.
predicate_uri	TEXT	required	Ontology fact, e.g. lab approved, bag created, lot charged.
object_uri	TEXT	—	Ontology object when it is another URI.
object_literal	TEXT	—	Ontology object when it is a value (KG, bag code, status).
object_datatype	VARCHAR(128)	—	Literal datatype if used.
source_table	VARCHAR(128)	—	Which MES table produced this triple.
source_id	UUID	—	Source row id.
export_status	VARCHAR(32)	required	Whether the triple was exported from staging.
created_at	TIMESTAMP	required	When the row was created.
updated_at	TIMESTAMP	required	When the row was last changed.
workspace_id	UUID	FK	Which workspace / plant area this row belongs to.